ADVANCED EXCEL · LESSON 9 OF 16

Date, Time & Number Functions and Named Ranges

Work confidently with dates and times (DATE, EOMONTH, EDATE, WORKDAY, NETWORKDAYS, WEEKDAY, WEEKNUM, DATEDIF, fiscal quarters), clean up numbers with INT, MOD, ROUND, CEILING and FLOOR, and make formulas readable with static and dynamic named ranges and LET.

● Live classes 7–8 ✦ Excel for Microsoft 365, 2026 Intermediate ⏱ 120 min

By the end of this lesson you can

- Explain how Excel stores dates and times, and do date arithmetic correctly
- Extract parts of dates, find month ends, add months, and count working days with holidays
- Calculate age, tenure and fiscal-year quarters (Indian April–March and US October–September)
- Use INT, MOD, ROUND, ROUNDUP, ROUNDDOWN, MROUND, CEILING.MATH, FLOOR.MATH, SQRT, POWER and ABS
- Create static and dynamic named ranges and simplify long formulas with LET

Datasets used

`superstore_sales.csv` `employees_small.csv`

Dates and times are numbers

Excel stores a date as the number of days since **1 January 1900** (that day is 1). A time is a **fraction of a day**: 12:00 noon is 0.5 and 06:00 is 0.25. A date with a time is simply a whole number plus a fraction.

What is really stored

	A	B	C
1	**You see**	**Stored as**	**Explanation**
2	27-04-2026	46139	Days since 1-Jan-1900
3	12:00	0.5	Half a day
4	06:30	0.2708	6.5 hours ÷ 24
5	27-04-2026 18:00	46139.75	Date + time

```
B2    =A2
B3    =TIME(12,0,0)
B4    =TIME(6,30,0)
B5    =DATE(2026,4,27)+TIME(18,0,0)
```

This is why date arithmetic works:

Date arithmetic

	A	B	C	D	E
1	**Order date**	**Ship date**	**Days to ship =B2-A2**	**Due date (+30 days)**	**Overdue? (as of 15-Jun-2026)**
2	27-04-2026	02-05-2026	5	27-05-2026	Overdue
3	20-05-2026	21-05-2026	1	19-06-2026	OK
4	01-06-2026	08-06-2026	7	01-07-2026	OK

```
C2    =B2-A2
D2    =A2+30
E2    =IF(DATE(2026,6,15)>D2,"Overdue","OK")
```

Each pattern is copied down/across to the other cells in its column.

TIP **TODAY() and NOW()**

`=TODAY()` returns today's date and `=NOW()` the date and time. They are **volatile**: they recalculate every time the workbook changes, so results move every day. For reports that must not change, type the "as of" date into a cell and refer to it, as in the overdue example above. Shortcuts: `Ctrl + ;` types today's date as a fixed value.

Building and taking apart dates

Function	Returns	Example (A2 = 27-04-2026)	Result
`DATE(y, m, d)`	A date from parts	`=DATE(2026,4,27)`	27-04-2026
`YEAR`, `MONTH`, `DAY`	Parts of a date	`=MONTH(A2)`	4
`WEEKDAY(date, 2)`	1 = Monday … 7 = Sunday	`=WEEKDAY(A2,2)`	1
`WEEKNUM(date, 2)`	Week of the year (weeks start Monday)	`=WEEKNUM(A2,2)`	18
`ISOWEEKNUM(date)`	ISO 8601 week number	`=ISOWEEKNUM(A2)`	18
`TEXT(date, "mmm")`	Month name as text	`=TEXT(A2,"mmmm")`	April
`EOMONTH(date, n)`	Last day of the month n months away	`=EOMONTH(A2,0)`	30-04-2026
`EDATE(date, n)`	Same day n months later	`=EDATE(A2,3)`	27-07-2026
`DATEVALUE(text)`	Converts date text to a real date	`=DATEVALUE("27-Apr-2026")`	46139

Date parts and month boundaries

	A	B	C	D	E	F	G	H
1	Date	Year	Month	Day	Weekday (Mon=1)	Month end	First of month	+3 month
2	27-04-2026	2026	4	27	1	30-04-2026	01-04-2026	27-07-202
3	31-01-2026	2026	1	31	6	31-01-2026	01-01-2026	28-02-202
4	29-02-2024	2024	2	29	4	29-02-2024	01-02-2024	28-02-202

```
B2   =YEAR(A2)
C2   =MONTH(A2)
D2   =DAY(A2)
E2   =WEEKDAY(A2,2)
F2   =EOMONTH(A2,0)
G2   =EOMONTH(A2,-1)+1
H2   =EDATE(A2,3)
```

Each pattern is copied down/across to the other cells in its column.

Notice that EDATE handles month ends correctly: 31-Jan + 1 month = 28-Feb (2026 is not a leap year), and 29-Feb-2024 + 12 months = 28-Feb-2025.

Working days

Function	Use
`NETWORKDAYS(start, end, [holidays])`	Count Mon–Fri days between two dates (inclusive) minus holidays
`NETWORKDAYS.INTL(start, end, weekend, [holidays])`	Choose weekend days: 11 = Sunday only, `"0000011"` = Sat + Sun
`WORKDAY(start, days, [holidays])`	The date after adding n working days
`WORKDAY.INTL(start, days, weekend, [holidays])`	Same with a custom weekend

SLA and delivery planning (holidays in G2:G4)

	A	B	C	D	E
1	Ticket	Opened	Closed	Working days =NETWORKDAYS(B2,C2,$G$2:$G$4)	Due (+5 working days)
2	T-101	10-08-2026	18-08-2026	7	17-08-2026
3	T-102	16-10-2026	23-10-2026	5	26-10-2026
4	T-103	06-11-2026	10-11-2026	4	12-11-2026

```
D2   =NETWORKDAYS(B2,C2,$G$2:$G$4)
E2   =WORKDAY(B2,5,$G$2:$G$4)
D3   =NETWORKDAYS(B3,C3,$G$2:$G$4)
E3   =WORKDAY(B3,5,$G$2:$G$4)
D4   =NETWORKDAYS.INTL(B4,C4,11,$G$2:$G$4)
E4   =WORKDAY.INTL(B4,5,11,$G$2:$G$4)
```

T-103 uses a **Sunday-only weekend** (code 11), common in Indian factories and retail. The holiday on 15-Aug-2026 falls on a Saturday, so it does not change T-101's count of working days.

Age, tenure and durations

`DATEDIF(start, end, unit)` is a hidden function (no tooltip) that returns complete years, months or days between two dates.

Unit	Returns
`"y"`	Complete years
`"m"`	Complete months
`"d"`	Days
`"ym"`	Months after removing complete years
`"md"`	Days after removing complete months

Tenure as of 31-Mar-2026 (in F1)

	A	B	C	D	E	F
1	**Employee**	**Join date**	**Years**	**Months after years**	**Tenure label**	31-03-2026
2	Kavya Iyer	28-02-2016	10	1	10 yrs 1 mths	
3	Imran Reddy	14-12-2019	6	3	6 yrs 3 mths	
4	Saanvi Rao	18-02-2025	1	1	1 yrs 1 mths	

```
C2   =DATEDIF(B2,$F$1,"y")
D2   =DATEDIF(B2,$F$1,"ym")
E2   =C2&" yrs "&D2&" mths"
```

Each pattern is copied down/across to the other cells in its column.

(Cell F1 holds the date 31-03-2026.) For fractional years use `=YEARFRAC(B2,$F$1)`, for example 10.09 years.

Fiscal years and quarters

Calendar quarter: `="Q"&ROUNDUP(MONTH(A2)/3,0)`. Many organisations use a different financial year:

Year type	Q1	Formula for the quarter number
Calendar	Jan–Mar	`=ROUNDUP(MONTH(A2)/3,0)`
India (FY Apr–Mar)	Apr–Jun	`=ROUNDUP(MOD(MONTH(A2)-4,12)/3+0.01,0)` or `=CHOOSE(MONTH(A2),4,4,4,1,1,1,2,2,2,3,3,3)`
US federal (FY Oct–Sep)	Oct–Dec	`=CHOOSE(MONTH(A2),2,2,2,3,3,3,4,4,4,1,1,1)`

Calendar, Indian FY and US FY quarters

	A	B	C	D	E
1	Date	Calendar Q	India FY quarter	India FY label	US FY quarter
2	10-02-2026	Q1	Q4	FY26	Q2
3	27-04-2026	Q2	Q1	FY27	Q3
4	03-11-2026	Q4	Q3	FY27	Q1

```
B2   ="Q"&ROUNDUP(MONTH(A2)/3,0)
C2   ="Q"&CHOOSE(MONTH(A2),4,4,4,1,1,1,2,2,2,3,3,3)
D2   ="FY"&(YEAR(A2)+(MONTH(A2)≥4))-2000
```

Each pattern is copied down/across to the other cells in its column.

In the Indian convention, 27-Apr-2026 falls in **FY27** (April 2026 – March 2027), quarter Q1.

Times

Function	Example	Result
`TIME(h, m, s)`	`=TIME(14,30,0)`	14:30
`HOUR`, `MINUTE`, `SECOND`	`=HOUR(A2)`	14
Duration in hours	`=(end-start)*24`	8.5
Duration beyond 24 h	Format `[h]:mm`	37:30
Shifts past midnight	`=MOD(end-start,1)`	Works when end is the next day

A very common clean-up task: a system exports times as plain numbers like `1430` meaning 14:30. Convert them with `INT` and `MOD`:

Convert 1430-style numbers into real times

	A	B	C	D	E
1	Raw	Hours =INT(A2/100)	Minutes =MOD(A2,100)	Time =TIME(INT(A2/100),MOD(A2,100),0)	Shift lengt
2	930	9	30	09:30	
3	1430	14	30	14:30	
4	2245	22	45	22:45	
5	5	0	5	00:05	
6	1815	18	15	18:15	

```
B2   =INT(A2/100)
C2   =MOD(A2,100)
D2   =TIME(INT(A2/100),MOD(A2,100),0)
E2   =MOD(D6-D2,1)*24
```

Each pattern is copied down/across to the other cells in its column.

Number functions

Function	What it does	Example	Result
`INT(n)`	Rounds **down** to the integer	`=INT(-7.6)`	`-8`
`TRUNC(n, [d])`	Cuts off decimals (towards zero)	`=TRUNC(-7.6)`	`-7`
`ROUND(n, d)`	Normal rounding	`=ROUND(7.456,2)`	7.46
`ROUNDUP(n, d)`	Always away from zero	`=ROUNDUP(7.401,0)`	8
`ROUNDDOWN(n, d)`	Always towards zero	`=ROUNDDOWN(7.99,0)`	7
`ROUND(n, -3)`	Round to thousands	`=ROUND(123456,-3)`	123000
`MROUND(n, m)`	Nearest multiple	`=MROUND(17,5)`	15
`CEILING.MATH(n, m)`	Up to the next multiple	`=CEILING.MATH(23,7)`	28
`FLOOR.MATH(n, m)`	Down to the previous multiple	`=FLOOR.MATH(23,7)`	21
`MOD(n, d)`	Remainder	`=MOD(28,3)`	1
`QUOTIENT(n, d)`	Integer part of a division	`=QUOTIENT(28,3)`	9
`ABS(n)`	Absolute value	`=ABS(-450)`	450
`SIGN(n)`	−1, 0 or 1	`=SIGN(-450)`	`-1`
`SQRT(n)`	Square root	`=SQRT(144)`	12
`POWER(n, p)`	Power (same as ^)	`=POWER(25,2)`	625

Rounding functions side by side

	A	B	C	D	E	F	G	
1	Value	INT	ROUND 0	ROUNDUP 0	ROUNDDOWN 0	ROUND 2	CEILING.MATH to 5	FLOOR.I
2	7.456	7	7	8	7	7.46	10	
3	-7.456	-8	-7	-8	-7	-7.46	-5	
4	23.5	23	24	24	23	23.5	25	

```
B2   =INT(A2)
C2   =ROUND(A2,0)
D2   =ROUNDUP(A2,0)
E2   =ROUNDDOWN(A2,0)
G2   =CEILING.MATH(A2,5)
H2   =FLOOR.MATH(A2,5)
```

Each pattern is copied down/across to the other cells in its column.

Practical uses

- **Packaging**: boxes needed for 523 items at 24 per box = `=CEILING.MATH(523,24)/24` or `=ROUNDUP(523/24,0)` → 22.
- **Price points**: round prices up to the next 99: `=CEILING.MATH(A2,100)-1` turns 1,234 into 1,299.
- **Split profit and loss** into two columns with `ABS`:

Profit and Loss columns

	A	B	C	D
1	**Store**	**Profit / Loss**	**Profit**	**Loss**
2	Pune	45200	45200	0
3	Delhi	-12800	0	12800
4	Kochi	3300	3300	0
5	Patna	-7600	0	7600
6	**Total**	**28100**	**48500**	**20400**

```
C2   =IF(B2>0,B2,0)
D2   =IF(B2<0,ABS(B2),0)
B6   =SUM(B2:B5)
C6   =SUM(C2:C5)
D6   =SUM(D2:D5)
```

Each pattern is copied down/across to the other cells in its column.

Named ranges

A **name** is a label for a cell, a range, a constant or a formula. `=SUM(Sales)` is easier to read and audit than `=SUM(Data!$O$2:$O$6324)`.

Static named ranges

Method	How
Name Box	Select the range, click the Name Box (left of the formula bar), type `GST_Rate`, press Enter
Define Name	Formulas → Define Name: set the name, scope (workbook or one sheet) and comment
Create from Selection	Select a table with headers → Formulas → Create from Selection (`Ctrl + Shift + F3`) → Top row: one name per column
Name Manager	Formulas → Name Manager (`Ctrl + F3`) to edit, delete or find broken names

Rules: names start with a letter or underscore, contain no spaces (use _), cannot look like a cell address (`Q1` or `FY26` are not allowed) and are not case-sensitive.

Formulas with names (GST_Rate = F1, Qty and Price are named columns)

	A	B	C	D	E	F
1	Qty	Price	Amount =Qty*Price	With GST =Qty*Price*(1+GST_Rate)		18%
2	4	1250	5000	5,900.00		
3	10	299	2990	3,528.20		
4	2	8999	17998	21,237.64		

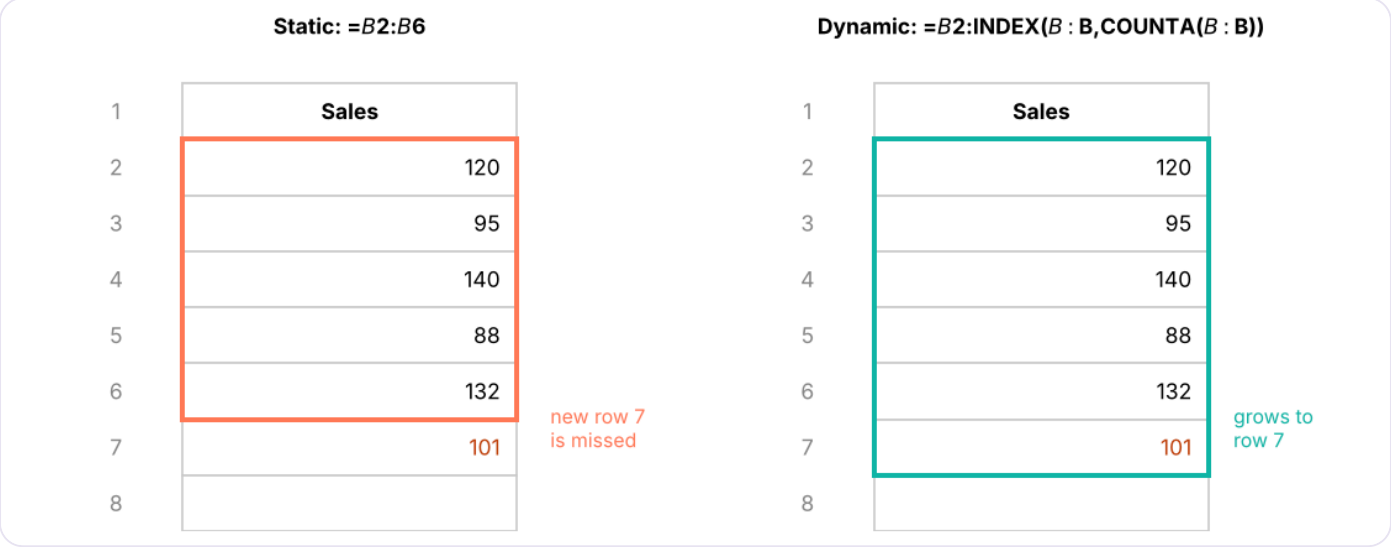
```
C2   =A2*B2
D2   =A2*B2*(1+$F$1)
C3   =A3*B3
D3   =A3*B3*(1+$F$1)
C4   =A4*B4
D4   =A4*B4*(1+$F$1)
```

(F1, the yellow cell, is named `GST_Rate`. With names defined, the formula in D2 reads `=Qty*Price*(1+GST_Rate)`.)

Dynamic named ranges

A static name like `=$B$2:$B$100` does not grow when you add row 101. A **dynamic** named range expands automatically, which keeps charts, validation lists and formulas up to date.

Method	Refers to	Notes
Excel Table (best)	`=tblSales[Amount]`	Automatic, no formula needed
`OFFSET + COUNTA`	`=OFFSET(Data!$B$2,0,0,COUNTA(Data!$B:$B)-1,1)`	Classic; volatile (recalculates often)
`INDEX + COUNTA`	`=Data!$B$2:INDEX(Data!$B:$B,COUNTA(Data!$B:$B))`	Not volatile; preferred over OFFSET
Spill reference	`=Data!$H$2#`	Points to a dynamic-array result



A static range stays fixed; a dynamic range grows with the data.

LET: names inside one formula

LET (Microsoft 365) assigns names to values **inside** a formula. It makes long formulas readable and faster, because each part is calculated once.

Without LET:
`=IF(SUMIFS(F:F,C:C,"Finance")/COUNTIFS(C:C,"Finance")>90000,"High","Normal")`

With LET:

Excel formula

```
=LET(
    total, SUMIFS(F:F, C:C, "Finance"),
    heads, COUNTIFS(C:C, "Finance"),
    avg,   total / heads,
    IF(avg > 90000, "High", "Normal")
)
```

LAMBDA goes one step further and lets you create your own reusable functions; see Lesson 12.

Case: date analysis on Superstore

How long do orders take to ship, by ship mode, and which weekday gets the most orders? In Excel you would add columns `=ship_date-order_date` and `=TEXT(order_date,"ddd")` and pivot them. Here are the real answers:

Python

```
import pandas as pd
s = pd.read_csv("superstore_sales.csv", parse_dates=["order_date", "ship_date"])
orders = s.drop_duplicates("order_id").copy()
orders["days_to_ship"] = (orders.ship_date - orders.order_date).dt.days            # =C2-B2
orders["weekday"] = orders.order_date.dt.day_name().str[:3]                         #
=TEXT(B2,"ddd")
orders["fy"] = "FY" + (orders.order_date.dt.year + (orders.order_date.dt.month ≥
4)).astype(str).str[2:]   # Indian FY
print(orders.groupby("ship_mode")["days_to_ship"].agg(["mean", "min",
"max"]).round(1).to_string())
print()
print(orders.groupby("fy").size().rename("orders").to_string())
```

OUTPUT

```
                mean  min  max
ship_mode
First Class      1.5    1    2
Same Day         0.0    0    0
Second Class     3.0    2    4
Standard Class   5.5    4    7

fy
FY23     169
FY24     952
FY25    1114
FY26     965
```

Summary

KEY TAKEAWAYS

- Dates are whole numbers (days since 1-Jan-1900); times are fractions of a day. Subtract dates to get days; multiply time differences by 24 for hours.
- `EOMONTH` and `EDATE` handle month ends correctly; `NETWORKDAYS` and `WORKDAY` (with `.INTL` for custom weekends) handle working days and holidays.
- `DATEDIF` gives complete years/months; build fiscal quarters with `CHOOSE(MONTH(…),…)`.
- `INT` rounds down, `TRUNC` cuts towards zero, `ROUNDUP`/`ROUNDDOWN` force direction, `CEILING.MATH`/`FLOOR.MATH` round to multiples, `MOD` gives remainders.
- Named ranges make formulas readable; use Tables or `INDEX`-based names for dynamic ranges, and `LET` for names inside a formula.

Quick quiz

6 questions · tap an answer to check it

1. A1 contains 27-04-2026 and B1 contains 02-05-2026. What does `=B1-A1` return?

A 5 days as text

B 5

C 0.05

D **VALUE!**

Answer: B. Dates are numbers, so subtracting gives the number of days.

2. Which formula returns the last day of the month for the date in A2?

A `=DATE(YEAR(A2),MONTH(A2),31)`

B `=EOMONTH(A2,0)`

C `=EDATE(A2,1)-1`

D `=MONTH(A2)+30`

Answer: B. EOMONTH(date,0) returns the month end; the other options fail for 30-day months or February.

3. What does `=INT(-3.2)` return?

A −3

B −4

C 3

D −3.2

Answer: B. INT always rounds down (towards minus infinity). TRUNC(-3.2) would give −3.

4. Which formula converts the number 1745 into the time 17:45?

A `=TIME(1745)`

B `=1745/24`

C `=TIME(INT(A2/100),MOD(A2,100),0)`

D `=TEXT(1745,"hh:mm")`

Answer: C. INT(1745/100) = 17 hours and MOD(1745,100) = 45 minutes.

5. Why is an INDEX-based dynamic named range often preferred to OFFSET?

- **A** It works only in Excel 365
- **B** OFFSET is volatile and recalculates on every change, which can slow large workbooks
- **C** INDEX cannot refer to whole columns
- **D** OFFSET does not grow with the data

Answer: B. Both grow, but OFFSET is volatile.

6. In the Indian financial year, which quarter is 15-January?

- **A** Q1
- **B** Q2
- **C** Q3
- **D** Q4

Answer: D. FY runs April–March, so January–March is Q4.

EXERCISE **HR date calculations**

Using `employees_small.csv` and an "as of" date of 31-03-2026 in a cell: (1) tenure in complete years; (2) the probation end date (6 months after joining); (3) the first working day after the probation end; (4) the joining quarter in the Indian FY.

Show solution (worked answer)

With join date in E2 and the as-of date in $J$1: (1) `=DATEDIF(E2,$J$1,"y")` (2) `=EDATE(E2,6)` (3) `=WORKDAY(EDATE(E2,6)-1,1)` (the first working day on or after the probation end) (4) `="Q"&CHOOSE(MONTH(E2),4,4,4,1,1,1,2,2,2,3,3,3)`

INTERVIEW CORNER

Questions you may be asked

Q: How does Excel store dates and times? Dates as serial numbers counting days from 1-Jan-1900; times as fractions of a day. This makes arithmetic, sorting and formatting consistent.

Q: How do you count working days between two dates excluding holidays? `NETWORKDAYS(start, end, holidays)` or `NETWORKDAYS.INTL` with a custom weekend code.

Q: What is a dynamic named range and when do you need one? A name whose range grows with the data, built with an Excel Table, `INDEX/COUNTA` or `OFFSET/COUNTA`. It keeps charts, validation lists and formulas current as data is added.

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